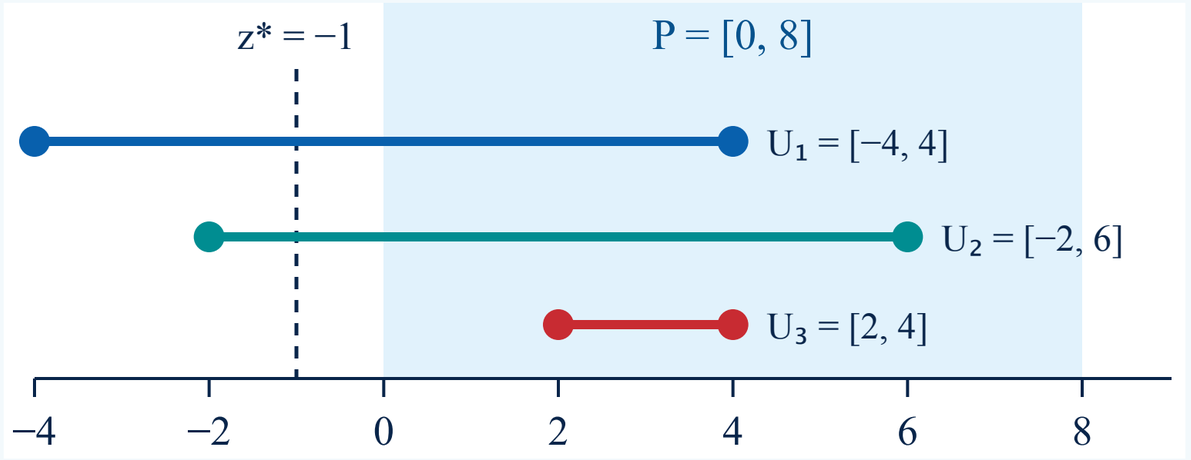
Example parameters
$\Omega = [-4, 8] \cap \mathbb{Z}$, $P = [0, 8] \cap \mathbb{Z}$, $(n, f, r, k) = (3, 1, 0, 2)$

1 Signed evidence

$U_1 = [-4, 4]$ Honest

$U_2 = [-2, 6]$ Honest

$U_3 = [2, 4]$ Byzantine

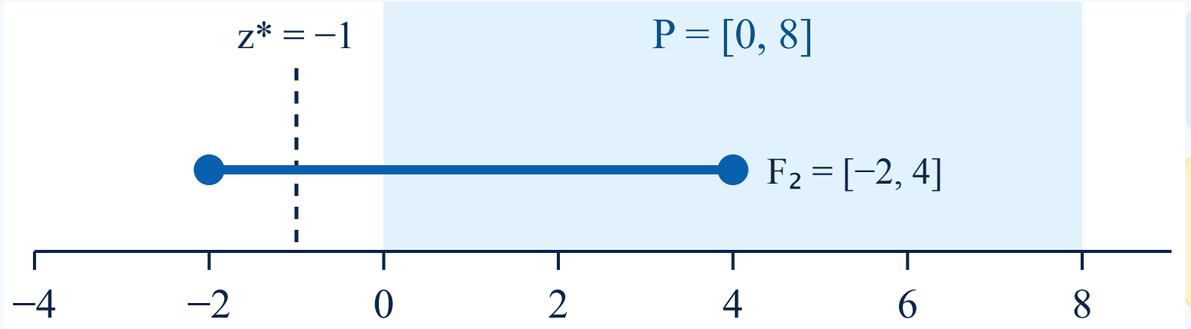


Key observation

Both honest intervals cover the true state $z^* = -1$; the Byzantine interval excludes it.

2 Threshold support

$F_2 = [-2, 4]$
supported by ≥ 2 issuers

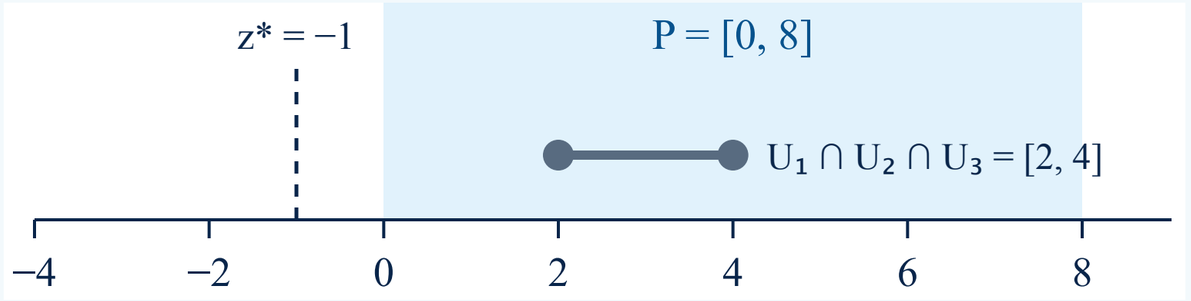


Supported by at least $k = 2$ issuers.

F_2 intersects inside and outside P → Ambiguous.

3 All-source overlap

$U_1 \cap U_2 \cap U_3 = [2, 4]$
intersection of all sources



Entirely inside P; would falsely accept if using all-source intersection alone.